

Bandwidth & Storage Estimates for Shielded Outputs

Hathor Network — Issue #1603 PoC (per-device extrapolation)

Methodology

Per-(N,M) total transaction payload (bytes) is read from results_memory/total_payload.csv produced by benchmark_memory.py. For each device we hold a typical real-world cellular and Wi-Fi throughput (Mbps) and a typical free-storage budget (GB).

Upload/download time is computed as $(\text{bytes} \times 8) / (\text{Mbps} \times 10^6)$; storage slots as $(\text{free_GB} \times 2^{30}) / \text{payload_bytes}$. The reference Intel i5-11300H row is shown only for the ratio context — the device has no cellular modem.

What goes in the payload

Per shielded output: 33 B Pedersen commitment + 33 B blinded asset generator + ~4.8 KiB Borromean range proof + ~70–140 B asset surjection proof. The range proof DOMINATES — 96–98% of the payload at any N. The surjection proof grows ~linearly with N up to about $\log_2(N)$ used inputs (libsecp caps the used-input set).

Typical vs peak

Each time cell shows TWO values. The upper bold value is the "typical" case (Opensignal/Speedtest median for that modem class). The italicized lower value is the "peak" case (best-case burst on good signal, mmWave where available, unshared Wi-Fi channel); generally 5–10× faster. The cell background color is driven by the TYPICAL time. For ranking we compute the 50/50 mix between typical and peak (weighted "realistic" view).

Caveats

The 5–10× peak/typical swing follows Opensignal's published distributions; real variance can be wider in either direction. Storage budgets assume the most-common SKU after OS + moderate apps. The reference baseline range proof was sized with amount=10¹⁸ (≈60-bit) — a tiny amount like 500 would shrink the proof to ~750 B because libsecp uses min_bits=0 (auto-min-encoding).

Time-cell color legend



Shielded Payload Size — diagonal of N×M grid

Reference values from results_memory/total_payload.csv. Per-output ≈ 4.9 KiB (Borromean range proof dominates).

(N=M)	Total payload	Per output	Range proof / out	Surjection / out (avg)
1	4.83 KiB	4.83 KiB	4.70 KiB	67 B
2	9.72 KiB	4.86 KiB	4.70 KiB	99 B
4	19.57 KiB	4.89 KiB	4.70 KiB	131 B
8	39.15 KiB	4.89 KiB	4.70 KiB	131 B
16	78.31 KiB	4.89 KiB	4.70 KiB	132 B
32	156.69 KiB	4.90 KiB	4.70 KiB	134 B
64	313.62 KiB	4.90 KiB	4.70 KiB	138 B

Cellular UPLOAD time per shielded transaction

Upper bold = typical; lower *italic* = peak (best-case burst). Color follows the typical time.

Device	Category	UL Mbps typ / peak	(N=M=1) typ / peak	(N=M=2) typ / peak	(N=M=4) typ / peak	(N=M=8) typ / peak	(N=M=16) typ / peak	(N=M=32) typ / peak	(N=M=64) typ / peak
Intel Core i5-11300H (reference)	Reference (this PoC)	—	— —	— —	— —	— —	— —	— —	— —
iPhone 16 Pro Max	Most powerful	60 / 220	660 μs <i>180 μs</i>	1.3 ms <i>362 μs</i>	2.7 ms <i>729 μs</i>	5.3 ms <i>1.5 ms</i>	10.7 ms <i>2.9 ms</i>	21.4 ms <i>5.8 ms</i>	42.8 ms <i>11.7 ms</i>
Samsung Galaxy S25 Ultra	Most powerful		720 μs <i>198 μs</i>	1.4 ms <i>398 μs</i>	2.9 ms <i>802 μs</i>	5.8 ms <i>1.6 ms</i>	11.7 ms <i>3.2 ms</i>	23.3 ms <i>6.4 ms</i>	46.7 ms <i>12.8 ms</i>
OnePlus 13	Most powerful		792 μs <i>220 μs</i>	1.6 ms <i>443 μs</i>	3.2 ms <i>891 μs</i>	6.4 ms <i>1.8 ms</i>	12.8 ms <i>3.6 ms</i>	25.7 ms <i>7.1 ms</i>	51.4 ms <i>14.3 ms</i>
iPhone 15	USA top-3	40 / 150	989 μs <i>264 μs</i>	2.0 ms <i>531 μs</i>	4.0 ms <i>1.1 ms</i>	8.0 ms <i>2.1 ms</i>	16.0 ms <i>4.3 ms</i>	32.1 ms <i>8.6 ms</i>	64.2 ms <i>17.1 ms</i>
iPhone 14	USA + Europe top-3	35 / 120	1.1 ms <i>330 μs</i>	2.3 ms <i>664 μs</i>	4.6 ms <i>1.3 ms</i>	9.2 ms <i>2.7 ms</i>	18.3 ms <i>5.3 ms</i>	36.7 ms <i>10.7 ms</i>	73.4 ms <i>21.4 ms</i>
Samsung Galaxy S23	USA top-3	40 / 150	989 μs <i>264 μs</i>	2.0 ms <i>531 μs</i>	4.0 ms <i>1.1 ms</i>	8.0 ms <i>2.1 ms</i>	16.0 ms <i>4.3 ms</i>	32.1 ms <i>8.6 ms</i>	64.2 ms <i>17.1 ms</i>
Samsung Galaxy A54	Europe + ROW top-3	20 / 80	2.0 ms <i>495 μs</i>	4.0 ms <i>996 μs</i>	8.0 ms <i>2.0 ms</i>	16.0 ms <i>4.0 ms</i>	32.1 ms <i>8.0 ms</i>	64.2 ms <i>16.0 ms</i>	128.5 ms <i>32.1 ms</i>
Xiaomi Redmi Note 12	Europe + ROW top-3	12 / 50	3.3 ms <i>792 μs</i>	6.6 ms <i>1.6 ms</i>	13.4 ms <i>3.2 ms</i>	26.7 ms <i>6.4 ms</i>	53.5 ms <i>12.8 ms</i>	107.0 ms <i>25.7 ms</i>	214.1 ms <i>51.4 ms</i>
Samsung Galaxy A14	Rest of World top-3	8 / 30	4.9 ms <i>1.3 ms</i>	10.0 ms <i>2.7 ms</i>	20.0 ms <i>5.3 ms</i>	40.1 ms <i>10.7 ms</i>	80.2 ms <i>21.4 ms</i>	160.4 ms <i>42.8 ms</i>	321.2 ms <i>85.6 ms</i>
Tecno Spark 10	Rest of World top-3	6 / 25	6.6 ms <i>1.6 ms</i>	13.3 ms <i>3.2 ms</i>	26.7 ms <i>6.4 ms</i>	53.5 ms <i>12.8 ms</i>	106.9 ms <i>25.7 ms</i>	213.9 ms <i>51.3 ms</i>	428.2 ms <i>102.8 ms</i>
iPhone 6 (2014)	2014 era	3 / 25	13.2 ms <i>1.6 ms</i>	26.6 ms <i>3.2 ms</i>	53.5 ms <i>6.4 ms</i>	106.9 ms <i>12.8 ms</i>	213.8 ms <i>25.7 ms</i>	427.9 ms <i>51.3 ms</i>	856.4 ms <i>102.8 ms</i>
Samsung Galaxy S5 (2014)	2014 era	2 / 20	19.8 ms <i>2.0 ms</i>	39.8 ms <i>4.0 ms</i>	80.2 ms <i>8.0 ms</i>	160.4 ms <i>16.0 ms</i>	320.8 ms <i>32.1 ms</i>	641.8 ms <i>64.2 ms</i>	1.28 s <i>128.5 ms</i>
LG G3 (2014)	2014 era	2 / 20	19.8 ms <i>2.0 ms</i>	39.8 ms <i>4.0 ms</i>	80.2 ms <i>8.0 ms</i>	160.4 ms <i>16.0 ms</i>	320.8 ms <i>32.1 ms</i>	641.8 ms <i>64.2 ms</i>	1.28 s <i>128.5 ms</i>

Cellular DOWNLOAD time per shielded transaction

Upper bold = typical; lower italic = peak. Peak cellular DL benefits most from mmWave 5G.

Device	Category	DL Mbps typ / peak	(N=M=1) typ / peak	(N=M=2) typ / peak	(N=M=4) typ / peak	(N=M=8) typ / peak	(N=M=16) typ / peak	(N=M=32) typ / peak	(N=M=64) typ / peak
Intel Core i5-11300H (reference)	Reference (this PoC)	—	— —	— —	— —	— —	— —	— —	— —
iPhone 16 Pro Max	Most powerful	250 / 1800	158 μs <i>22 μs</i>	319 μs <i>44 μs</i>	641 μs <i>89 μs</i>	1.3 ms <i>178 μs</i>	2.6 ms <i>356 μs</i>	5.1 ms <i>713 μs</i>	10.3 ms <i>1.4 ms</i>
Samsung Galaxy S25 Ultra	Most powerful		172 μs <i>26 μs</i>	346 μs <i>53 μs</i>	697 μs <i>107 μs</i>	1.4 ms <i>214 μs</i>	2.8 ms <i>428 μs</i>	5.6 ms <i>856 μs</i>	11.2 ms <i>1.7 ms</i>
OnePlus 13	Most powerful		180 μs <i>33 μs</i>	362 μs <i>66 μs</i>	729 μs <i>134 μs</i>	1.5 ms <i>267 μs</i>	2.9 ms <i>535 μs</i>	5.8 ms <i>1.1 ms</i>	11.7 ms <i>2.1 ms</i>
iPhone 15	USA top-3	180 / 1200	220 μs <i>33 μs</i>	443 μs <i>66 μs</i>	891 μs <i>134 μs</i>	1.8 ms <i>267 μs</i>	3.6 ms <i>535 μs</i>	7.1 ms <i>1.1 ms</i>	14.3 ms <i>2.1 ms</i>
iPhone 14	USA + Europe top-3	150 / 900	264 μs <i>44 μs</i>	531 μs <i>89 μs</i>	1.1 ms <i>178 μs</i>	2.1 ms <i>356 μs</i>	4.3 ms <i>713 μs</i>	8.6 ms <i>1.4 ms</i>	17.1 ms <i>2.9 ms</i>
Samsung Galaxy S23	USA top-3	170 / 1000	233 μs <i>40 μs</i>	469 μs <i>80 μs</i>	943 μs <i>160 μs</i>	1.9 ms <i>321 μs</i>	3.8 ms <i>642 μs</i>	7.6 ms <i>1.3 ms</i>	15.1 ms <i>2.6 ms</i>
Samsung Galaxy A54	Europe + ROW top-3	90 / 500	440 μs <i>79 μs</i>	885 μs <i>159 μs</i>	1.8 ms <i>321 μs</i>	3.6 ms <i>641 μs</i>	7.1 ms <i>1.3 ms</i>	14.3 ms <i>2.6 ms</i>	28.5 ms <i>5.1 ms</i>
Xiaomi Redmi Note 12	Europe + ROW top-3	50 / 300	792 μs <i>132 μs</i>	1.6 ms <i>266 μs</i>	3.2 ms <i>535 μs</i>	6.4 ms <i>1.1 ms</i>	12.8 ms <i>2.1 ms</i>	25.7 ms <i>4.3 ms</i>	51.4 ms <i>8.6 ms</i>
Samsung Galaxy A14	Rest of World top-3	25 / 150	1.6 ms <i>264 μs</i>	3.2 ms <i>531 μs</i>	6.4 ms <i>1.1 ms</i>	12.8 ms <i>2.1 ms</i>	25.7 ms <i>4.3 ms</i>	51.3 ms <i>8.6 ms</i>	102.8 ms <i>17.1 ms</i>
Tecno Spark 10	Rest of World top-3	20 / 120	2.0 ms <i>330 μs</i>	4.0 ms <i>664 μs</i>	8.0 ms <i>1.3 ms</i>	16.0 ms <i>2.7 ms</i>	32.1 ms <i>5.3 ms</i>	64.2 ms <i>10.7 ms</i>	128.5 ms <i>21.4 ms</i>
iPhone 6 (2014)	2014 era	8 / 60	4.9 ms <i>660 μs</i>	10.0 ms <i>1.3 ms</i>	20.0 ms <i>2.7 ms</i>	40.1 ms <i>5.3 ms</i>	80.2 ms <i>10.7 ms</i>	160.4 ms <i>21.4 ms</i>	321.2 ms <i>42.8 ms</i>
Samsung Galaxy S5 (2014)	2014 era	6 / 50	6.6 ms <i>792 μs</i>	13.3 ms <i>1.6 ms</i>	26.7 ms <i>3.2 ms</i>	53.5 ms <i>6.4 ms</i>	106.9 ms <i>12.8 ms</i>	213.9 ms <i>25.7 ms</i>	428.2 ms <i>51.4 ms</i>
LG G3 (2014)	2014 era	6 / 50	6.6 ms <i>792 μs</i>	13.3 ms <i>1.6 ms</i>	26.7 ms <i>3.2 ms</i>	53.5 ms <i>6.4 ms</i>	106.9 ms <i>12.8 ms</i>	213.9 ms <i>25.7 ms</i>	428.2 ms <i>51.4 ms</i>

Wi-Fi UPLOAD time per shielded transaction

Upper bold = typical sustained; lower *italic* = peak burst (unshared channel, short range).

Device	Category	UL Mbps typ / peak	(N=M=1) typ / peak	(N=M=2) typ / peak	(N=M=4) typ / peak	(N=M=8) typ / peak	(N=M=16) typ / peak	(N=M=32) typ / peak	(N=M=64) typ / peak
Intel Core i5-11300H (reference)	Reference (this PoC)	300 / 700	132 μ s <i>57 μs</i>	266 μ s <i>114 μs</i>	535 μ s <i>229 μs</i>	1.1 ms <i>458 μs</i>	2.1 ms <i>916 μs</i>	4.3 ms <i>1.8 ms</i>	8.6 ms <i>3.7 ms</i>
iPhone 16 Pro Max	Most powerful	700 / 1600	57 μ s <i>25 μs</i>	114 μ s <i>50 μs</i>	229 μ s <i>100 μs</i>	458 μ s <i>200 μs</i>	916 μ s <i>401 μs</i>	1.8 ms <i>802 μs</i>	3.7 ms <i>1.6 ms</i>
Samsung Galaxy S25 Ultra	Most powerful		49 μ s <i>18 μs</i>	100 μ s <i>36 μs</i>	200 μ s <i>73 μs</i>	401 μ s <i>146 μs</i>	802 μ s <i>292 μs</i>	1.6 ms <i>583 μs</i>	3.2 ms <i>1.2 ms</i>
OnePlus 13	Most powerful		49 μ s <i>18 μs</i>	100 μ s <i>36 μs</i>	200 μ s <i>73 μs</i>	401 μ s <i>146 μs</i>	802 μ s <i>292 μs</i>	1.6 ms <i>583 μs</i>	3.2 ms <i>1.2 ms</i>
iPhone 15	USA top-3	500 / 1200	79 μ s <i>33 μs</i>	159 μ s <i>66 μs</i>	321 μ s <i>134 μs</i>	641 μ s <i>267 μs</i>	1.3 ms <i>535 μs</i>	2.6 ms <i>1.1 ms</i>	5.1 ms <i>2.1 ms</i>
iPhone 14	USA + Europe top-3	400 / 800	99 μ s <i>49 μs</i>	199 μ s <i>100 μs</i>	401 μ s <i>200 μs</i>	802 μ s <i>401 μs</i>	1.6 ms <i>802 μs</i>	3.2 ms <i>1.6 ms</i>	6.4 ms <i>3.2 ms</i>
Samsung Galaxy S23	USA top-3	600 / 1200	66 μ s <i>33 μs</i>	133 μ s <i>66 μs</i>	267 μ s <i>134 μs</i>	535 μ s <i>267 μs</i>	1.1 ms <i>535 μs</i>	2.1 ms <i>1.1 ms</i>	4.3 ms <i>2.1 ms</i>
Samsung Galaxy A54	Europe + ROW top-3	250 / 600	158 μ s <i>66 μs</i>	319 μ s <i>133 μs</i>	641 μ s <i>267 μs</i>	1.3 ms <i>535 μs</i>	2.6 ms <i>1.1 ms</i>	5.1 ms <i>2.1 ms</i>	10.3 ms <i>4.3 ms</i>
Xiaomi Redmi Note 12	Europe + ROW top-3	200 / 433	198 μ s <i>91 μs</i>	398 μ s <i>184 μs</i>	802 μ s <i>370 μs</i>	1.6 ms <i>741 μs</i>	3.2 ms <i>1.5 ms</i>	6.4 ms <i>3.0 ms</i>	12.8 ms <i>5.9 ms</i>
Samsung Galaxy A14	Rest of World top-3	80 / 200	495 μ s <i>198 μs</i>	996 μ s <i>398 μs</i>	2.0 ms <i>802 μs</i>	4.0 ms <i>1.6 ms</i>	8.0 ms <i>3.2 ms</i>	16.0 ms <i>6.4 ms</i>	32.1 ms <i>12.8 ms</i>
Tecno Spark 10	Rest of World top-3	60 / 150	660 μ s <i>264 μs</i>	1.3 ms <i>531 μs</i>	2.7 ms <i>1.1 ms</i>	5.3 ms <i>2.1 ms</i>	10.7 ms <i>4.3 ms</i>	21.4 ms <i>8.6 ms</i>	42.8 ms <i>17.1 ms</i>
iPhone 6 (2014)	2014 era	30 / 80	1.3 ms <i>495 μs</i>	2.7 ms <i>996 μs</i>	5.3 ms <i>2.0 ms</i>	10.7 ms <i>4.0 ms</i>	21.4 ms <i>8.0 ms</i>	42.8 ms <i>16.0 ms</i>	85.6 ms <i>32.1 ms</i>
Samsung Galaxy S5 (2014)	2014 era	20 / 60	2.0 ms <i>660 μs</i>	4.0 ms <i>1.3 ms</i>	8.0 ms <i>2.7 ms</i>	16.0 ms <i>5.3 ms</i>	32.1 ms <i>10.7 ms</i>	64.2 ms <i>21.4 ms</i>	128.5 ms <i>42.8 ms</i>
LG G3 (2014)	2014 era	20 / 60	2.0 ms <i>660 μs</i>	4.0 ms <i>1.3 ms</i>	8.0 ms <i>2.7 ms</i>	16.0 ms <i>5.3 ms</i>	32.1 ms <i>10.7 ms</i>	64.2 ms <i>21.4 ms</i>	128.5 ms <i>42.8 ms</i>

Wi-Fi DOWNLOAD time per shielded transaction

Upper bold = typical sustained; lower *italic* = peak burst.

Device	Category	DL Mbps typ / peak	(N=M=1) typ / peak	(N=M=2) typ / peak	(N=M=4) typ / peak	(N=M=8) typ / peak	(N=M=16) typ / peak	(N=M=32) typ / peak	(N=M=64) typ / peak
Intel Core i5-11300H (reference)	Reference (this PoC)	900 / 1800	44 μ s <i>22 μs</i>	89 μ s <i>44 μs</i>	178 μ s <i>89 μs</i>	356 μ s <i>178 μs</i>	713 μ s <i>356 μs</i>	1.4 ms <i>713 μs</i>	2.9 ms <i>1.4 ms</i>
iPhone 16 Pro Max	Most powerful	1500 / 3500	26 μ s <i>11 μs</i>	53 μ s <i>23 μs</i>	107 μ s <i>46 μs</i>	214 μ s <i>92 μs</i>	428 μ s <i>183 μs</i>	856 μ s <i>367 μs</i>	1.7 ms <i>734 μs</i>
Samsung Galaxy S25 Ultra	Most powerful	1800 / 4800	22 μ s <i>8 μs</i>	44 μ s <i>17 μs</i>	89 μ s <i>33 μs</i>	178 μ s <i>67 μs</i>	356 μ s <i>134 μs</i>	713 μ s <i>267 μs</i>	1.4 ms <i>535 μs</i>
OnePlus 13	Most powerful	1800 / 4800	22 μ s <i>8 μs</i>	44 μ s <i>17 μs</i>	89 μ s <i>33 μs</i>	178 μ s <i>67 μs</i>	356 μ s <i>134 μs</i>	713 μ s <i>267 μs</i>	1.4 ms <i>535 μs</i>
iPhone 15	USA top-3	900 / 2400	44 μ s <i>16 μs</i>	89 μ s <i>33 μs</i>	178 μ s <i>67 μs</i>	356 μ s <i>134 μs</i>	713 μ s <i>267 μs</i>	1.4 ms <i>535 μs</i>	2.9 ms <i>1.1 ms</i>
iPhone 14	USA + Europe top-3	600 / 1200	66 μ s <i>33 μs</i>	133 μ s <i>66 μs</i>	267 μ s <i>134 μs</i>	535 μ s <i>267 μs</i>	1.1 ms <i>535 μs</i>	2.1 ms <i>1.1 ms</i>	4.3 ms <i>2.1 ms</i>
Samsung Galaxy S23	USA top-3	1200 / 2400	33 μ s <i>16 μs</i>	66 μ s <i>33 μs</i>	134 μ s <i>67 μs</i>	267 μ s <i>134 μs</i>	535 μ s <i>267 μs</i>	1.1 ms <i>535 μs</i>	2.1 ms <i>1.1 ms</i>
Samsung Galaxy A54	Europe + ROW top-3	400 / 1200	99 μ s <i>33 μs</i>	199 μ s <i>66 μs</i>	401 μ s <i>134 μs</i>	802 μ s <i>267 μs</i>	1.6 ms <i>535 μs</i>	3.2 ms <i>1.1 ms</i>	6.4 ms <i>2.1 ms</i>
Xiaomi Redmi Note 12	Europe + ROW top-3	300 / 866	132 μ s <i>46 μs</i>	266 μ s <i>92 μs</i>	535 μ s <i>185 μs</i>	1.1 ms <i>370 μs</i>	2.1 ms <i>741 μs</i>	4.3 ms <i>1.5 ms</i>	8.6 ms <i>3.0 ms</i>
Samsung Galaxy A14	Rest of World top-3	150 / 433	264 μ s <i>91 μs</i>	531 μ s <i>184 μs</i>	1.1 ms <i>370 μs</i>	2.1 ms <i>741 μs</i>	4.3 ms <i>1.5 ms</i>	8.6 ms <i>3.0 ms</i>	17.1 ms <i>5.9 ms</i>
Tecno Spark 10	Rest of World top-3	120 / 300	330 μ s <i>132 μs</i>	664 μ s <i>266 μs</i>	1.3 ms <i>535 μs</i>	2.7 ms <i>1.1 ms</i>	5.3 ms <i>2.1 ms</i>	10.7 ms <i>4.3 ms</i>	21.4 ms <i>8.6 ms</i>
iPhone 6 (2014)	2014 era	50 / 150	792 μ s <i>264 μs</i>	1.6 ms <i>531 μs</i>	3.2 ms <i>1.1 ms</i>	6.4 ms <i>2.1 ms</i>	12.8 ms <i>4.3 ms</i>	25.7 ms <i>8.6 ms</i>	51.4 ms <i>17.1 ms</i>
Samsung Galaxy S5 (2014)	2014 era	40 / 120	989 μ s <i>330 μs</i>	2.0 ms <i>664 μs</i>	4.0 ms <i>1.3 ms</i>	8.0 ms <i>2.7 ms</i>	16.0 ms <i>5.3 ms</i>	32.1 ms <i>10.7 ms</i>	64.2 ms <i>21.4 ms</i>
LG G3 (2014)	2014 era	40 / 120	989 μ s <i>330 μs</i>	2.0 ms <i>664 μs</i>	4.0 ms <i>1.3 ms</i>	8.0 ms <i>2.7 ms</i>	16.0 ms <i>5.3 ms</i>	32.1 ms <i>10.7 ms</i>	64.2 ms <i>21.4 ms</i>

Storage Capacity — Shielded Transactions per Free-Space Budget

Slots = (Free GB / total bytes per transaction). Free GB is a typical budget for the most-common SKU after OS + apps.

Device	Category	Free GB	MB/s	(N=M=1) slots	(N=M=2) slots	(N=M=4) slots	(N=M=8) slots	(N=M=16) slots	(N=M=32) slots	(N=M=64) slots
Intel Core i5-11300H (reference)	Reference (this PoC)	400	3000	86.8M	43.1M	21.4M	10.7M	5.4M	2.7M	1.3M
iPhone 16 Pro Max	Most powerful	220	3500	47.8M	23.7M	11.8M	5.9M	2.9M	1.5M	735.5k
Samsung Galaxy S25 Ultra	Most powerful	220	4000	47.8M	23.7M	11.8M	5.9M	2.9M	1.5M	735.5k
OnePlus 13	Most powerful	220	4000	47.8M	23.7M	11.8M	5.9M	2.9M	1.5M	735.5k
iPhone 15	USA top-3	100	1500	21.7M	10.8M	5.4M	2.7M	1.3M	669.2k	334.3k
iPhone 14	USA + Europe top-3	100	1500	21.7M	10.8M	5.4M	2.7M	1.3M	669.2k	334.3k
Samsung Galaxy S23	USA top-3	180	2900	39.1M	19.4M	9.6M	4.8M	2.4M	1.2M	601.8k
Samsung Galaxy A54	Europe + ROW top-3	90	500	19.5M	9.7M	4.8M	2.4M	1.2M	602.3k	300.9k
Xiaomi Redmi Note 12	Europe + ROW top-3	90	400	19.5M	9.7M	4.8M	2.4M	1.2M	602.3k	300.9k
Samsung Galaxy A14	Rest of World top-3	45	200	9.8M	4.9M	2.4M	1.2M	602.5k	301.1k	150.5k
Tecno Spark 10	Rest of World top-3	90	200	19.5M	9.7M	4.8M	2.4M	1.2M	602.3k	300.9k
iPhone 6 (2014)	2014 era	8	50	1.7M	862.6k	428.6k	214.3k	107.1k	53.5k	26.7k
Samsung Galaxy S5 (2014)	2014 era	8	40	1.7M	862.6k	428.6k	214.3k	107.1k	53.5k	26.7k
LG G3 (2014)	2014 era	12	40	2.6M	1.3M	642.8k	321.4k	160.7k	80.3k	40.1k

Devices ranked by TYPICAL cellular UL+DL at N=M=16

Median real-world throughput · payload 78.31 KiB

#	Device	Category	Value	SoC
1	iPhone 16 Pro Max	Most powerful	10.7 ms UL + 2.6 ms DL = 13.3 ms	Apple A18 Pro
2	Samsung Galaxy S25 Ultra	Most powerful	11.7 ms UL + 2.8 ms DL = 14.5 ms	Snapdragon 8 Elite
3	OnePlus 13	Most powerful	12.8 ms UL + 2.9 ms DL = 15.7 ms	Snapdragon 8 Elite
4	iPhone 15	USA top-3	16.0 ms UL + 3.6 ms DL = 19.6 ms	Apple A16 Bionic
5	Samsung Galaxy S23	USA top-3	16.0 ms UL + 3.8 ms DL = 19.8 ms	Snapdragon 8 Gen 2
6	iPhone 14	USA + Europe top-3	18.3 ms UL + 4.3 ms DL = 22.6 ms	Apple A15 Bionic
7	Samsung Galaxy A54	Europe + ROW top-3	32.1 ms UL + 7.1 ms DL = 39.2 ms	Exynos 1380
8	Xiaomi Redmi Note 12	Europe + ROW top-3	53.5 ms UL + 12.8 ms DL = 66.3 ms	Snapdragon 685
9	Samsung Galaxy A14	Rest of World top-3	80.2 ms UL + 25.7 ms DL = 105.9 ms	MediaTek Helio G80
10	Tecno Spark 10	Rest of World top-3	106.9 ms UL + 32.1 ms DL = 139.0 ms	MediaTek Helio G88
11	iPhone 6 (2014)	2014 era	213.8 ms UL + 80.2 ms DL = 294.0 ms	Apple A8
12	Samsung Galaxy S5 (2014)	2014 era	320.8 ms UL + 106.9 ms DL = 427.7 ms	Snapdragon 801
13	LG G3 (2014)	2014 era	320.8 ms UL + 106.9 ms DL = 427.7 ms	Snapdragon 801

Devices ranked by PEAK cellular UL+DL at N=M=16

Best-case burst (5G mmWave etc.) · payload 78.31 KiB

#	Device	Category	Value	SoC
1	iPhone 16 Pro Max	Most powerful	2.9 ms UL + 356 μs DL = 3.3 ms	Apple A18 Pro
2	Samsung Galaxy S25 Ultra	Most powerful	3.2 ms UL + 428 μs DL = 3.6 ms	Snapdragon 8 Elite
3	OnePlus 13	Most powerful	3.6 ms UL + 535 μs DL = 4.1 ms	Snapdragon 8 Elite
4	iPhone 15	USA top-3	4.3 ms UL + 535 μs DL = 4.8 ms	Apple A16 Bionic
5	Samsung Galaxy S23	USA top-3	4.3 ms UL + 642 μs DL = 4.9 ms	Snapdragon 8 Gen 2
6	iPhone 14	USA + Europe top-3	5.3 ms UL + 713 μs DL = 6.1 ms	Apple A15 Bionic
7	Samsung Galaxy A54	Europe + ROW top-3	8.0 ms UL + 1.3 ms DL = 9.3 ms	Exynos 1380
8	Xiaomi Redmi Note 12	Europe + ROW top-3	12.8 ms UL + 2.1 ms DL = 15.0 ms	Snapdragon 685
9	Samsung Galaxy A14	Rest of World top-3	21.4 ms UL + 4.3 ms DL = 25.7 ms	MediaTek Helio G80
10	Tecno Spark 10	Rest of World top-3	25.7 ms UL + 5.3 ms DL = 31.0 ms	MediaTek Helio G88
11	iPhone 6 (2014)	2014 era	25.7 ms UL + 10.7 ms DL = 36.4 ms	Apple A8
12	Samsung Galaxy S5 (2014)	2014 era	32.1 ms UL + 12.8 ms DL = 44.9 ms	Snapdragon 801
13	LG G3 (2014)	2014 era	32.1 ms UL + 12.8 ms DL = 44.9 ms	Snapdragon 801

Devices ranked by BLENDED cellular UL+DL at N=M=16

50/50 typical+peak — realistic mixed-condition view · payload 78.31 KiB

#	Device	Category	Value	SoC
1	iPhone 16 Pro Max	Most powerful	typ 13.3 ms · peak 3.3 ms · blend 8.3 ms	Apple A18 Pro
2	Samsung Galaxy S25 Ultra	Most powerful	typ 14.5 ms · peak 3.6 ms · blend 9.0 ms	Snapdragon 8 Elite
3	OnePlus 13	Most powerful	typ 15.7 ms · peak 4.1 ms · blend 9.9 ms	Snapdragon 8 Elite
4	iPhone 15	USA top-3	typ 19.6 ms · peak 4.8 ms · blend 12.2 ms	Apple A16 Bionic
5	Samsung Galaxy S23	USA top-3	typ 19.8 ms · peak 4.9 ms · blend 12.4 ms	Snapdragon 8 Gen 2
6	iPhone 14	USA + Europe top-3	typ 22.6 ms · peak 6.1 ms · blend 14.3 ms	Apple A15 Bionic
7	Samsung Galaxy A54	Europe + ROW top-3	typ 39.2 ms · peak 9.3 ms · blend 24.3 ms	Exynos 1380
8	Xiaomi Redmi Note 12	Europe + ROW top-3	typ 66.3 ms · peak 15.0 ms · blend 40.6 ms	Snapdragon 685
9	Samsung Galaxy A14	Rest of World top-3	typ 105.9 ms · peak 25.7 ms · blend 65.8 ms	MediaTek Helio G80
10	Tecno Spark 10	Rest of World top-3	typ 139.0 ms · peak 31.0 ms · blend 85.0 ms	MediaTek Helio G88
11	iPhone 6 (2014)	2014 era	typ 294.0 ms · peak 36.4 ms · blend 165.2 ms	Apple A8
12	Samsung Galaxy S5 (2014)	2014 era	typ 427.7 ms · peak 44.9 ms · blend 236.3 ms	Snapdragon 801
13	LG G3 (2014)	2014 era	typ 427.7 ms · peak 44.9 ms · blend 236.3 ms	Snapdragon 801

Devices ranked by STORAGE CAPACITY at N=M=16 (78.31 KiB payload)

More tx slots in the free-space budget is better; flash write time is sub-millisecond on every device.

#	Device	Category	Value	SoC
1	iPhone 16 Pro Max	Most powerful	2.9M slots, write 22 μs	Apple A18 Pro
2	Samsung Galaxy S25 Ultra	Most powerful	2.9M slots, write 19 μs	Snapdragon 8 Elite
3	OnePlus 13	Most powerful	2.9M slots, write 19 μs	Snapdragon 8 Elite
4	Samsung Galaxy S23	USA top-3	2.4M slots, write 26 μs	Snapdragon 8 Gen 2
5	iPhone 15	USA top-3	1.3M slots, write 51 μs	Apple A16 Bionic
6	iPhone 14	USA + Europe top-3	1.3M slots, write 51 μs	Apple A15 Bionic
7	Samsung Galaxy A54	Europe + ROW top-3	1.2M slots, write 153 μs	Exynos 1380
8	Xiaomi Redmi Note 12	Europe + ROW top-3	1.2M slots, write 191 μs	Snapdragon 685
9	Tecno Spark 10	Rest of World top-3	1.2M slots, write 382 μs	MediaTek Helio G88
10	Samsung Galaxy A14	Rest of World top-3	602.5k slots, write 382 μs	MediaTek Helio G80
11	LG G3 (2014)	2014 era	160.7k slots, write 1.9 ms	Snapdragon 801
12	iPhone 6 (2014)	2014 era	107.1k slots, write 1.5 ms	Apple A8
13	Samsung Galaxy S5 (2014)	2014 era	107.1k slots, write 1.9 ms	Snapdragon 801

Sources & References

Cellular throughput (typical median real-world)

Opensignal global mobile network experience reports: <https://www.opensignal.com/reports>
Speedtest Global Index (Ookla): <https://www.speedtest.net/global-index>
Per-modem-class typical UL/DL distilled from the device-class chapters of the above.

Wi-Fi throughput (typical fast home router)

Wi-Fi 7 / 6E / 6 / 5 / 4 device support taken from each device's GSMArena spec sheet.
Throughputs assume an unshared 5 GHz / 6 GHz channel at close range – a generous upper bound for what the modem class can sustain in practice.

Storage budget

Most-common SKU per device (e.g. iPhone 16 Pro Max $\approx$ 256 GB $\rightarrow$ $\approx$ 220 GB free after iOS + apps).
Apple specs: <https://support.apple.com/specs>
Samsung specs: <https://www.samsung.com/global/galaxy/>
Generic OEM specs: <https://www.gsmarena.com/>

Flash sequential write speed

UFS 4.0 (S25 Ultra): $\sim$ 3500 MB/s – JEDEC UFS 4.0 spec
UFS 3.1 (S23, OnePlus): $\sim$ 1500 MB/s
Apple NVMe (iPhone 14+): $\sim$ 1500 MB/s sustained sequential write
eMMC 5.1 (mid-range): 150–500 MB/s
eMMC 4.5 (entry/2014): 25–50 MB/s

Payload sizes

Empirically measured by benchmark_memory.py (this repo) using
hathor.crypto.shielded.* over the Rust hathor-ct-crypto library.
Range-proof primitive: secp256k1_rangeproof_sign (Borromean ring sigs),
NOT a Bulletproof – older comments in the repo are wrong.

Disclaimer

All numbers are illustrative. Throughput depends on network conditions
and routing; flash write speed depends on device thermal state and
flash wear; storage budgets vary widely by SKU. Treat as order-of-magnitude.